

HE2003 Econometrics I: Linear Regression with Multiple Regressors 4

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Tests and Confidence Intervals in Multiple Regression

KEY CONCEPT: The Chi-squared Distribution

The Chi-squared distribution with q degrees of freedom (denoted χ_q^2) is defined to be the distribution of the sum of q independent squared standard normal random variables. e.g., let Z_1, Z_2 , and Z_3 be independent $N(0, 1)$ random variables, then $Z_1^2 + Z_2^2 + Z_3^2$ has a Chi-squared distribution with 3 degrees of freedom.

Here, in large samples, F is distributed as χ_q^2/q (or denoted $F_{q,\infty}$).

Selected Critical Values of χ_q^2/q ($F_{q,\infty}$)

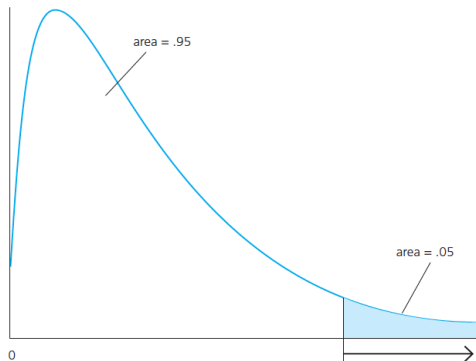
q	5% critical value
1	3.84 (why?)
2	3.00 (the case $q = 2$ in the example)
3	2.60
...	

Then, we can use these critical values to do the F-test with 5% level, just as we use 1.96 to do the t-test.

Tests and Confidence Intervals in Multiple Regression

Distribution of $\chi^2/q (F_{q,\infty})$

- The rejection region is at the right tail of the distribution.
- The 5% critical value is 3 for $F_{2,\infty}$, so we can reject at the 5% level when the computed $F > 3$; The 5% critical value is 2.6 for $F_{3,\infty}$, so we can reject at the 5% level when $F > 2.6$...



Tests and Confidence Intervals in Multiple Regression

Example: Parents' Education in a Birth Weight Equation (cont'd)

The **restricted regression** result (using `bwght_parentseduc.dta`) is:

$$\widehat{bwght} = 115.46 - 0.59cigs + 1.83parity + 0.06faminc, \quad R_r^2 = 0.036416.$$

And the **unrestricted regression** result is:

$$\begin{aligned} \widehat{bwght} = & 114.52 - 0.37motheduc + 0.47fatheduc - 0.59cigs \\ & + 1.78parity + 0.05faminc, \quad R_u^2 = 0.038748. \end{aligned}$$

In addition, $n = 1191$, $k = 5$, and $q = 2$ in this example. Thus, we obtain the value of the **Homoskedasticity-only F**:

$$F = \frac{(R_u^2 - R_r^2)/q}{(1 - R_u^2)/(n - k - 1)} = \frac{(0.038748 - 0.036416)/2}{(1 - 0.038748)/1185} = 1.44.$$

Tests and Confidence Intervals in Multiple Regression

STATA implementation with Homoskedasticity-only F-test

```
reg bwght cigs parity faminc
```

Source	SS	df	MS	Number of obs =	1191
Model	17579.8997	3	5859.96658	F(3, 1187) =	14.95
Residual	465166.792	1187	391.884408	Prob > F =	0.0000
				R-squared =	0.0364
				Adj R-squared =	0.0340
Total	482746.692	1190	405.669489	Root MSE =	19.796

bwght	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
cigs	-.5978519	.1087701	-5.50	0.000	-.8112549 -.3844489
parity	1.832274	.6575402	2.79	0.005	.5422035 3.122345
faminc	.0670618	.0323938	2.07	0.039	.0035063 .1306173
_cons	115.4699	1.655898	69.73	0.000	112.2211 118.7187

```
reg bwght motheduc fatheduc cigs parity faminc
```

Source	SS	df	MS	Number of obs =	1191
Model	18705.5567	5	3741.11135	F(5, 1185) =	9.55
Residual	464041.135	1185	391.595895	Prob > F =	0.0000
				R-squared =	0.0387
				Adj R-squared =	0.0347
Total	482746.692	1190	405.669489	Root MSE =	19.789

bwght	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
motheduc	-.3704503	.3198551	-1.16	0.247	-.9979957 .2570951
fatheduc	.4723944	.2826433	1.67	0.095	-.0821426 1.026931
cigs	-.5959362	.1103479	-5.40	0.000	-.8124352 -.3794373
parity	1.787603	.6594055	2.71	0.007	.4938709 3.081336
faminc	.0560414	.0365616	1.53	0.126	-.0156913 .1277742
_cons	114.5243	3.728453	30.72	0.000	107.2092 121.8394

Tests and Confidence Intervals in Multiple Regression

STATA also provides user-friendly command for doing F-test directly. For the **homoskedasticity-only F-test**, we can first run the regression with the **unrestricted model** (without the “robust” option). Then, we use the command “**test**”.

1. **reg bwght motheduc fatheduc cigs parity faminc**
2. **test motheduc fatheduc**

STATA reports the **F-statistic is equal to 1.44**, and its **p-value is equal to 0.238**. So we cannot reject $H_0 : \beta_1 = \beta_2 = 0$ (parents' education have no effect on baby's birthweight) at the 5% level.

```
. test motheduc fatheduc
```

```
( 1) motheduc = 0
```

```
( 2) fatheduc = 0
```

```
F( 2, 1185) =    1.44  
Prob > F =    0.2380
```

Tests and Confidence Intervals in Multiple Regression

The Heteroskedasticity-robust F-test

- The detailed formula for the heteroskedasticity-robust F-statistic is much more complicated than the homoskedasticity-only F-statistic, thus omitted.
- Although the formula is different, the heteroskedasticity-robust F-test still uses the critical values of χ^2/q ($F_{q,\infty}$). That is, the heteroskedasticity-robust F-test uses the same critical values as the homoskedasticity-only F-test.

The STATA commands for the heteroskedasticity-robust F is as follows (the “robust” option is needed here):

1. **reg bwght motheduc fatheduc cigs parity faminc, robust**
2. **test motheduc fatheduc**

Tests and Confidence Intervals in Multiple Regression

STATA implementatin with Heteroskedasticity-robust F-test

The **robust F-statistic** is equal to **1.64**, and its **p-value** is equal to **0.1944**.

Again, we cannot reject $H_0 : \beta_1 = \beta_2 = 0$ at the 5% level.

```
. reg bwght motheduc fatheduc cigs parity faminc, robust
```

Linear regression

```
Number of obs = 1191
F( 5, 1185) = 10.18
Prob > F      = 0.0000
R-squared     = 0.0387
Root MSE     = 19.789
```

bwght	Robust					
	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
motheduc	-.3704503	.2948507	-1.26	0.209	-.9489379	.2080374
fatheduc	.4723944	.2645678	1.79	0.074	-.0466791	.991468
cigs	-.5959362	.1069637	-5.57	0.000	-.8057955	-.386077
parity	1.787603	.6461293	2.77	0.006	.5199184	3.055288
faminc	.0560414	.0353185	1.59	0.113	-.0132523	.1253352
_cons	114.5243	3.499662	32.72	0.000	107.6581	121.3906

```
. test motheduc fatheduc
```

```
( 1) motheduc = 0
( 2) fatheduc = 0
```

```
F( 2, 1185) = 1.64
Prob > F = 0.1944
```

Tests and Confidence Intervals in Multiple Regression

KEY CONCEPT: The “overall” regression F -statistic

Often we want to test the joint hypothesis that **all the slope coefficients are zero**. That is, the (joint) null and alternative hypotheses are

$H_0 : \beta_1 = 0, \beta_2 = 0, \dots, \beta_k = 0$ vs. $H_1 : \beta_j \neq 0$, for at least one $j, j = 1, \dots, k$.

- Because R^2 is zero when there is no (explicit) regressor, $R_r^2 = 0$ in this case. In addition, here we have $q = k$, the **homo. F -statistic becomes**

$$F = \frac{R_u^2/k}{(1 - R_u^2)/(n - k - 1)}, \quad (1)$$

where R_u^2 is calculated from the unrestricted model.

- Example of birthweight and parents' education:**

$$F = (0.038748/5)/[(1 - 0.038748)/1185] = 9.55.$$

- Many statistical softwares such as STATA routinely report e.q.(1) when an OLS regression is estimated.

Tests and Confidence Intervals in Multiple Regression

Empirical Application: Testing the “Efficient Market Hypothesis” in Financial Economics by using the F -statistic (cont'd)

For 2019, let dkr be a firm's **debt to capital ratio**, let eps denote the **earning per share**, let $netinc$ denote the firm's **net income**, and let $salary$ denote the **compensation for CEO**.

The following equation was estimated:

$$\widehat{return} = -14.37 + 0.321dkr + 0.043eps - 0.0051netinc + 0.0035salary,$$

(6.89) (0.201) (0.078) (0.0047) (0.0022)

$n = 142, R^2 = 0.0395.$

Question: Test at the 5% significance level whether none of the regressors with information in 2019 are able to predict the stock return from 2019 to 2023. **Overall, is the evidence for predictability of stock returns strong or weak?**

Note: The 5% critical values for $F_{1,\infty}$, $F_{2,\infty}$, $F_{3,\infty}$, and $F_{4,\infty}$ are: 3.84, 3.00, 2.60, and 2.37, respectively.

Tests and Confidence Intervals in Multiple Regression

Summary: Testing Joint Null Hypothesis Using the F-statistics

- The “one at a time” approach of rejecting the null if one of the absolute values of t-statistics exceeds 1.96 **reject more than 5%** when the null is actually true (Type I error larger than 5%); Therefore, it **cannot be used to test joint null hypothesis**.
- The **homoskedasticity-only F-statistic** has a formula related to the R^2 of the restricted and unrestricted models: This F-statistic is comparing the fitting of two models;
- The **heteroskedasticity-robust F-statistic** uses the same critical values as the homoskedasticity-only F-statistic, and can be implemented in softwares such as STATA (using “robust” option in “reg” and then the command “test”).
- For large n , both F-statistics are **distributed χ_q^2/q (or called $F_{q,\infty}$)**.
(No need to recite the critical values for the F-test)

The Sveriges Riksbank Prize in
Economic Sciences in Memory of
Alfred Nobel 2021

David Card
Joshua D. Angrist
Guido W. Imbens

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11 October 2021

[The Royal Swedish Academy of Sciences](#) has decided to award the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2021

with one half to

David Card

University of California, Berkeley, USA

“for his empirical contributions to labour economics”

and the other half jointly to

Joshua D. Angrist

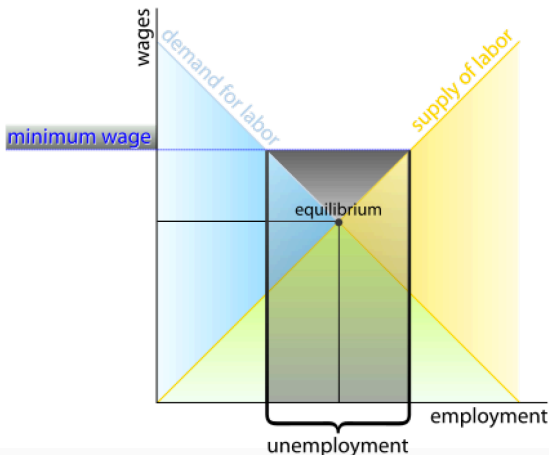
Massachusetts Institute of Technology, Cambridge, USA

Guido W. Imbens

Stanford University, USA

Interaction Regression and Difference-in-differences

Question: Will raising (minimum) wage help the poor? Many economists may doubt. **Standard economic theory:** in a competitive labor market, higher min.wage may **reduce employment, hurting the very workers** that policies were designed to help.

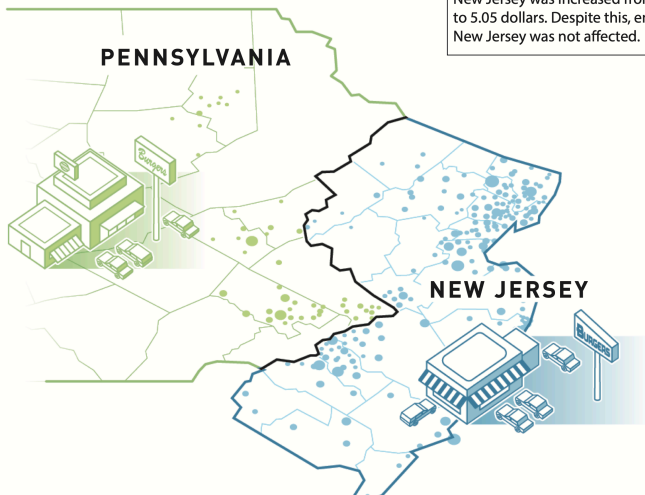


Interaction Regression and Difference-in-differences

Minimum wage change in New Jersey (Card and Krueger, American Economic Review, 1994)

● CONTROL GROUP

● TREATMENT GROUP



1 April 1992: The hourly minimum wage in New Jersey was increased from 4.25 dollars to 5.05 dollars. Despite this, employment in New Jersey was not affected.

Interaction Regression and Difference-in-differences

This is one of the first applications of difference-in-differences (DID).

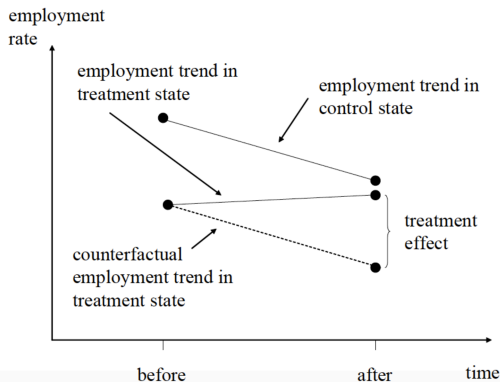
Minimum wage change in NJ on April 1, 1992:

- NJ raised the state minimum wage from \$4.25 to \$5.05
- The minimum wage in the neighbouring state Pennsylvania (PA) stayed at \$4.25
- Card and Krueger collected data on employment at fast food restaurants in NJ in Feb 1992 (before the raise) and again in Nov 1992 (after the raise)
- Data from the same type of restaurants in eastern PA
- What is the effect of employment? They compared the Feb-to-Nov change in employment in NJ to the change in employment in PA over the same period

Interaction Regression and Difference-in-differences

Question: Can we conclude that raising minimum wage increases the employment? Are there other factors which could lead to the increases in employment in NJ?

Economic growth with positive (or negative) macro trend or shock could result in the rising/decreasing employment in NJ. Then, how to control such macro trend?



Interaction Regression and Difference-in-differences

Let us define:

$\bar{Y}_{1,NJ}$: average employment in NJ before the policy change (Feb)

$\bar{Y}_{2,NJ}$: average employment in NJ after the policy change (Nov)

$\bar{Y}_{1,PA}$: average employment in PA before the policy change (Feb)

$\bar{Y}_{2,PA}$: average employment in PA after the policy change (Nov)

$\bar{Y}_{2,NJ} - \bar{Y}_{1,NJ}$: policy effect on the employment in NJ + employment time trend in NJ

$\bar{Y}_{2,PA} - \bar{Y}_{1,PA}$: employment time trend in PA

Here, the idea of the quasi-experiment is that without the change in minimum wage, NJ (treatment group) and PA (control group) would have had the **same time trend** in terms of employment during the period (which is called the **parallel trend assumption**).

	New Jersey	Pennsylvania	Difference
Before Increase	20.43	23.38	2.95
After Increase	20.90	21.10	0.20
Difference	0.47	-2.28	2.75

Interaction Regression and Difference-in-differences

After removing this trend, the change in employment in NJ should be regarded as the causal effect of minimum wage on employment:

$$\text{policy effect} = (\bar{Y}_{2,NJ} - \bar{Y}_{1,NJ}) - (\bar{Y}_{2,PA} - \bar{Y}_{1,PA}) = 2.75 \quad (2)$$

This is called the **difference-in-differences (DID) estimator**. Here, $\bar{Y}_{2,PA} - \bar{Y}_{1,PA}$ serves as the “counterfactual” for the treatment group NJ (what would the change in employment rate have been if NJ did not raise the min.wage).

Therefore, the estimation result shows that the employment in NJ was not negatively affected by the raise of the minimum wage! Card and Krueger's DID result shows that the standard economic theory may not hold in real world.

Question: How to construct the DID estimator by using a multiple regression model with dummy variables (and their interaction)?

Hint: We can define two dummy variables: One for NJ vs PA, and one for “Before” vs “After”.